

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0068

Exxonmobil Baton Rouge Turbine Generator 100 kV
May 02, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0068. The Interconnection Customer proposes a 74.5 MW battery energy storage facility interconnecting at the Exxonmobil Baton Rouge Turbine Generator 100 kV substation in the Ohio area. The request was received on September 07, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 56 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$39,390,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SWAN VLY-TWIN COUNTY RANCH 230.0 kV
Contingency: SWAN VLY SUBSTATION-TREGO 500.0 kV Line
Loading: 132.6% (Rating: 439 MVA)
DFAX: 14.8%
Pre-project loading: 130.0%
2. Facility: BEAVERLEY-GEISEL MONUMENT 100.0 kV
Contingency: EASTOWNRD 230/1kV Transformer Bank
Loading: 120.4% (Rating: 135 MVA)
DFAX: 16.5%
Pre-project loading: 111.3%
3. Facility: LENKURT-SOUTHWEST LIMA 100.0 kV
Contingency: EASTOWN RD 230 kV/1 kV Transformer
Loading: 119.6% (Rating: 153 MVA)
DFAX: 16.5%
Pre-project loading: 111.5%
4. Facility: SOUTHWEST LIMA-BEAVERLEY SUBSTATION 100.0 kV Line
Contingency: EASTOWN ROAD 230/1 kV Autotransformer
Loading: 116.0% (Rating: 149 MVA)
DFAX: 16.5%
Pre-project loading: 107.7%
5. Facility: E FINDLAY-11TH AVENUE N 100.0 kV
Contingency: EXXONMOBIL BATON ROUGE TURBINE GENERATOR-BUFFTON 100.0 kV Line
Loading: 104.7% (Rating: 147 MVA)
DFAX: 74.9%
Pre-project loading: 66.7%

6. Facility: BUFFTON-EXXONMOBIL BATON ROUGE TURBINE GENERATOR 100.0 kV
 Contingency: E FINDLAY-11TH AVENUE N 100.0 kV
 Loading: 103.2% (Rating: 131 MVA)
 DFAX: 81.3%
 Pre-project loading: 56.9%
7. Facility: 11TH AVENUE N-W MELROSE 100.0 kV Line 1
 Contingency: W MELROSE-11TH AVENUE N 100.0 kV Circuit 2
 Loading: 101.6% (Rating: 222 MVA)
 DFAX: 18.8%
 Pre-project loading: 95.3%
8. Facility: WEST MELROSE-11TH AVENUE N 100.0 kV Line 2
 Contingency: 11TH AVENUE N-WEST MELROSE 100.0 kV Ckt 1
 Loading: 100.1% (Rating: 225 MVA)
 DFAX: 18.8%
 Pre-project loading: 93.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SWAN VALLEY-TWIN COUNTY RANCH 230.0 kV	Rebuild line with higher-capacity conductor	\$25,660,000	100% (\$25,660,000)
BEAVERLEY-GEISEL MONUMENT 100.0 kV	Rebuild line with higher-capacity conductor	\$1,901,000	100% (\$1,901,000)
LENKURT-SOUTHWEST LIMA 100.0 kV	Rebuild line with higher-capacity conductor	\$5,160,000	100% (\$5,160,000)
SOUTHWEST LIMA-BEAVERLEY SUBSTATION 100.0 kV Line	Rebuild line with higher-capacity conductor	\$3,240,000	100% (\$3,240,000)
EAST FINDLAY-11TH AVENUE N 100.0 kV	Reconductor line	\$1,478,000	100% (\$1,478,000)
EXXONMOBIL BATON ROUGE TURBINE GENERATOR-BUFFTON 100.0 kV	Reconductor line	\$1,356,000	100% (\$1,356,000)
11TH AVENUE N SUBSTATION-W MELROSE 100.0 kV Line 1	Reconductor line	\$286,000	100% (\$286,000)
W MELROSE-11TH AVEUE N 100.0 kV #2	Reconductor line	\$309,000	100% (\$309,000)
	Total allocated to project		\$39,390,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.